

Identification of inrush currents in power transformers based on higher-order statistics



L.L. Zhang^a, Q.H. Wu^{a,b,*}, T.Y. Ji^a, A.Q. Zhang^a

^a School of Electric Power Engineering, South China University of Technology, Guangzhou 510641, China

^b Department of Electrical Engineering and Electronics, University of Liverpool, Liverpool L69 3GJ, UK

ARTICLE INFO

Article history:

Received 15 August 2016

Received in revised form 19 January 2017

Accepted 22 January 2017

Available online 1 February 2017

Keywords:

Higher-order statistics

Power transformer

Differential protection

Magnetizing inrush

Internal fault

ABSTRACT

This paper presents a novel method based on higher-order statistics for discrimination between magnetizing inrush and internal fault in power transformers. The distribution characteristics of the sample points of differential currents, under various operation conditions, are firstly investigated and quantified using higher-order statistics. Based on these characteristics, three kurtosis-based indices are defined for distinguishing inrush current from internal fault current. The final discrimination criterion combines these three indices to improve the performance of inrush current identification. Extensive simulation studies and experimental tests verify the effectiveness of the proposed method and its advantages over the conventional second harmonic restraint method.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

Differential relays are widely used for protecting power transformers from the damage caused by internal faults. However, energization of a transformer may result in large differential current which causes the operation of differential relay [1]. Therefore, it is necessary to discriminate magnetising inrush from internal fault condition and properly restrain the relay.

The second harmonic restraint method is the most widely used approach to inrush current identification in practical application. Another conventional method, i.e., the gap detection, uses the dwell time to identify inrush current. Some commercial differential relays combine these two methods to improve the reliability of inrush current identification, however, which may still maloperate in some cases, such as energizing a transformer with high remanence [2]. Therefore, a reliable inrush current identification method is much desired.

Numbers of techniques have been presented to address the inrush problem in the past decades. Improved harmonic-based methods were presented in [3,4]. The phase shift between primary and secondary currents of transformer was introduced for current discrimination in [5]. In [6], the magnitude of differential active power was used to discriminate inrush current from fault

current. An improved method based on the time-domain analyse of the differential power signal was presented in [7]. Compared to the conventional methods, these methods include the need of voltage transformers. Transformer model based method has also been reported [8], of which the performance relies on accurate parameter estimation. In other papers, inrush current and internal fault are directly recognized by the geometric features of the current waveform [9,10], or the correlation coefficient between waveforms [11,12].

Wavelet transform (WT) has been considered as an effective tool for transient analysis and thus has been widely employed to extract the features of differential current in power transformer [13–17]. In [18,19], inrush current is detected based on the correlation coefficients between energy vectors obtained by WT. Mathematical morphology (MM) has also been introduced to measure the difference between the waveforms of inrush current and fault current [20–22]. Nevertheless, these methods rely on the selection of a suitable basic function or structuring element. Advanced methods such as artificial neural network [23,24], decision making logic [25,26] and hidden Markov model [27] were also presented for inrush current discrimination. These advanced methods require a huge amount of historical data for training to obtain better performance, with the cost of higher computational load.

As a sequel to our earlier paper [28] which proposes a skewness-based differential protection for transmission lines, this paper attempts to reveal the difference between inrush current and fault current using higher-order statistics (HOS). This paper firstly

* Corresponding author at: School of Electric Power Engineering, South China University of Technology, Guangzhou 510641, China.

investigates the distribution characteristic of sample points of the currents under different operation conditions. Based on which, three indices are defined and their combination is used to discriminate inrush current from fault current. In order to evaluate the effectiveness of the proposed method, extensive simulation studies and experimental tests in laboratory were conducted. The results demonstrate that the proposed method can effectively identify the currents under various conditions, such as current transformer (CT) saturation, energization of transformer with high remanence, and energization of transformer with an internal fault.

2. Higher-order statistics based inrush identification

2.1. Distribution characteristic of sample points of different currents

Ideally, the current of a power transformer under normal operation contains only the fundamental frequency component. Fig. 1(a) shows a standard sinusoidal waveform with 80 sample points in a full fundamental frequency cycle. By binning the sample points into 20 equally spaced containers and counting the number of sample points in each container, we can depict the histogram shown in Fig. 1(b) as a graphical representation of the distribution of the current data. We can see that the histogram is symmetrical about the mean of all the sample points, i.e., 0 in this case.

When an internal fault occurs in the transformer, the fault current usually contains not only the fundamental frequency component but also a decaying DC component, as shown in Fig. 1(c). By the same way, we depict the histogram in Fig. 1(d) for studying the distribution of the sample points of the fault current. It is found that the distribution characteristic is similar to that of the current under normal operation, as the addition of decaying DC component has only a slight effect on the distribution of the sample points.

On the other hand, Fig. 1(e) shows the waveform of a typical inrush current, with all sample points located at one side of the

time axis. The distribution of the sample points is revealed by the histogram shown in Fig. 1(f), which is skewed to the right side and is obviously different from those of the currents under normal operation and the fault conditions.

Based on the above observations, we can know that an index can measure the difference between distribution characteristics will be suitable for discriminating transformer magnetizing inrush current from internal fault condition. Therefore, this paper employs higher-order statistics to quantify the distribution characteristic of the sample points of differential currents and presents a novel criterion for inrush identification.

2.2. Higher-order statistics

The higher-order statistics [29] refers to functions using the third or higher power of a sample, which are a recently developed class of tools that have shown promising potential in such fields as radar, image processing, seismology and array processing. HOS, such as *skewness* and *kurtosis*, are particularly used in estimation of shape parameters as well as measuring the deviation of a distribution from the normal distribution.

The skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable about its mean. The skewness of a random variable X is the moment coefficient of skewness. It is denoted by γ_3 and defined as:

$$\gamma_3 = E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] = \frac{\mu_3}{\sigma^3} \quad (1)$$

where μ_3 is the third central moment, μ is the mean, σ is the standard deviation, and E is the expectation operator.

The kurtosis is a descriptor of the shape of a probability distribution, which is the fourth standardized moment and is defined as:

$$\gamma_4 = \frac{E[(X - \mu)^4]}{(E[(X - \mu)^2])^2} = \frac{\mu_4}{\sigma^4} \quad (2)$$

The kurtosis of any univariate normal distribution is 3. It is common to compare the kurtosis of a distribution to this value. Distributions with kurtosis greater than 3 are said to be *leptokurtic* while distributions with kurtosis less than 3 are said to be *platykurtic*.

Under normal operation condition, the differential current of a power transformer has a sinusoidal waveform and the distribution of the sample points is symmetric about the mean, as shown in Fig. 1(a) and (b), of which the skewness is theoretically zero. In contrast, the distribution of the sample points of differential current under the condition of internal fault or inrush is asymmetric, which leads the skewness to be non-zero value. Therefore, internal fault and inrush can be treated as disturbance and can be detected by comparing the absolute value of the skewness to a threshold [28]. This paper focuses on the discrimination between inrush current and internal fault condition after the disturbance has been detected. The kurtosis can also be used to quantify the feature of differential current. In the following subsection, HOS-based indices are presented for inrush identification under different conditions.

2.3. HOS-based inrush identification indices

Suppose X is the data set with all the sample points of the differential current in a fundamental frequency cycle, of which the distribution characteristic can be evaluated by kurtosis, denoted by k . To reveal the local characteristic of differential currents under different conditions, we can separate the sample points into two subsets by their mean. One subset with the sample points whose values larger than the mean, denoted by X_1 , and another subset

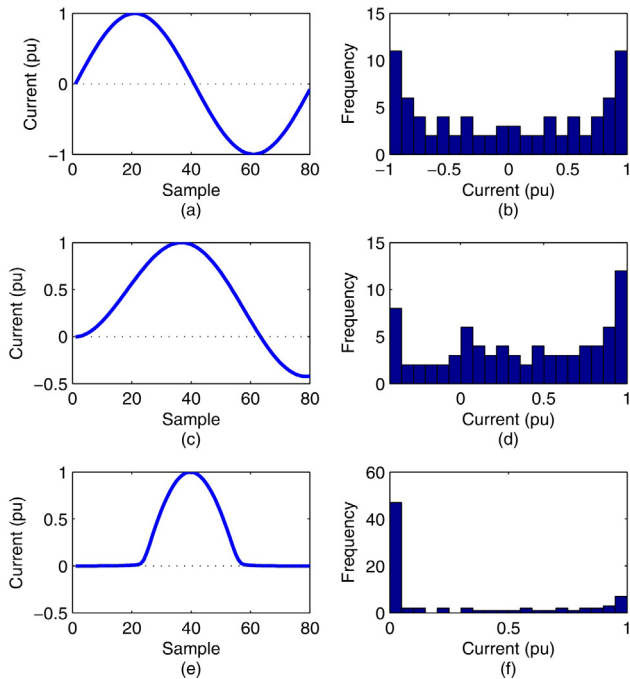


Fig. 1. Distributions of sample points of currents under different conditions: (a) Current under normal operation, (b) distribution of the sample points of current under normal operation, and (c) fault current, (d) distribution of the sample points of fault current, (e) inrush current, and (f) distribution of the sample points of inrush current.

Table 1
Kurtosis of the sample points of currents under different conditions.

Cases	k	k_1	k_2	k_3	k_4	$k_{\min}$	$k_{\max}$
Normal	1.5000	1.9128	1.9314	1.9498	1.9498	1.9128	1.9498
Fault	1.7554	1.9015	1.7337	1.9287	1.7641	1.7337	1.9287
Inrush	2.2308	2.0274	12.8797	1.4766	4.8392	1.4766	12.8797

with sample points whose values smaller than the mean, denoted by X_2 , of which the kurtosis are denoted by k_1 and k_2 , respectively. Consequently, we use k_1 to reveal the characteristic of the current around the peak of the current waveform and use k_2 to reveal the distribution characteristic of the sample points around the time axis or the valley.

The sample points can also be divided into two subsets by their median. The subset with sample points whose values larger than the median is denoted by X_3 and the subset with sample points whose values smaller than the median is denoted by X_4 , of which the kurtosis are denoted by k_3 and k_4 , respectively. Comparison studies are conducted to evaluate the distribution characteristic of the sample points of the three current waveforms in Fig. 1(a), (c) and (e). The results are summarized in Table 1, where $k_{\max}$ and $k_{\min}$ are the maximum value and minimum value of $\{k_1, k_2, k_3, k_4\}$, respectively.

The kurtosis of all sample points in a full fundamental frequency cycle of the three currents is 1.50, 1.7554 and 2.2308, respectively. As expected, the kurtosis for the case of fault current is close to that of the standard sinusoidal waveform whereas the kurtosis for an inrush current is much larger. Therefore, we can use the following index to identify inrush from internal fault:

$$K_1 = \frac{1.5}{k} \quad (3)$$

which compares the kurtosis of the sample points in a full fundamental frequency cycle of inrush current or fault current to that of a standard sinusoidal waveform, i.e., 1.5. Since k is larger than 1.5 for both of inrush current and fault current, K_1 is always smaller than 1. Moreover, K_1 is smaller for an inrush current than for a fault current.

From the last column in Table 1, we can also find that $k_{\max}$ is about 2 for both the cases of standard sinusoidal current and fault current while $k_{\max}$ is more than 12 for the inrush current. Consequently, the ratio between 2 and $k_{\max}$ of the differential current can be used as the second index, which is defined as:

$$K_2 = \frac{2}{k_{\max}} \quad (4)$$

By comparing the results in the last two columns in Table 1, we can find that the difference between $k_{\min}$ and $k_{\max}$ is much larger for inrush current than those of the other two cases. This means that the ratio between $k_{\min}$ and $k_{\max}$ can be used as the third index, which is defined as:

$$K_3 = \frac{k_{\min}}{k_{\max}} \quad (5)$$

These three indices can be used to individually distinguish inrush current from fault current in many cases, but the performance of them may decrease under different circumstances, such as CT saturation and symmetrical inrush current. Therefore, in order to well identify inrush currents from fault currents in all cases, the combination of these indices are employed as the final index, which is defined as:

$$K = 1 - cK_1K_2K_3 \quad (6)$$

where c is a compensation coefficient. For the first cycle after a disturbance is detected, $c = 1.3$, because the data window has not

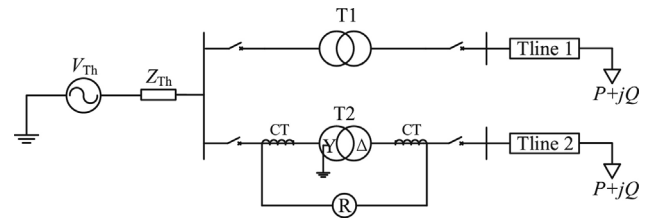


Fig. 2. The simulation model in PSCAD/EMTDC.

been fully filled with the sample points of inrush current or fault current yet. After that, $c = 1$.

As explained above, under fault conditions, K_1 , K_2 and K_3 are all about 1 and thus K is close to 0. In contrast, for an inrush current, K_1 , K_2 and K_3 are (much) smaller than 1, therefore, their product can be very close to 0 and K is about 1. Consequently, the HOS-based inrush identification criterion is defined as:

$$K > K_{\text{set}} \quad (7)$$

where K is calculated in a sliding data window with length of a full fundamental frequency cycle, and K_{set} is a pre-set threshold. If K is greater than the threshold, the proposed method identify the differential current as an inrush and restrain the operation of differential relay. On the other hand, if K is not greater than the threshold, the proposed method identify the differential current as a fault current and release the relay for operation.

3. Simulation studies

3.1. Simulation model

In order to evaluate the performance of the proposed method under various conditions, the power system shown in Fig. 2 is simulated using PSCAD/EMTDC. Most of the data and parameters of this power system are identical to those of the system used in [22]. The simulated transformer is a 380/110 kV, 100 MVA, Y/δ connected transformer. Different switching angles can be simulated by controlling the operating instant of breaker while different levels of remanence can be simulated by introducing controlled DC current sources [30].

For the purpose of comparison, the differential currents in all simulated cases were used to test both of the proposed method and the second harmonic restraint method. The threshold of the second harmonic restraint is set to be $R_{\text{set}} = 0.15$, which is widely used for power transformer protection in practice. In order to determine the threshold of the proposed method, 20 test cases of inrush conditions were firstly conducted with power transformer T2 switched at random instants. The differential currents of the power transformer in these 20 cases were recorded, considering CT saturation, and are measured using the proposed index K . The minimum value of K under different inrush conditions is about 0.5. In order to ensure the reliability, a coefficient of 0.5 is employed. Therefore, the threshold K_{set} is set to 0.25. For a different electrical system, one can determine the threshold in a similar way as described above.

A total of 105 cases were conducted to simulate inrush currents by energizing the transformer at different angles of voltage and with different levels of remanence. In addition, a total of 420 cases were conducted to simulate internal faults with different fault types, fault locations and fault inception instants. The system frequency is 50 Hz and the sampling frequency is 4 kHz. Besides, in order to verify the anti-noise ability of the proposed method, the original signals were added with 30 dB Gaussian white noise.

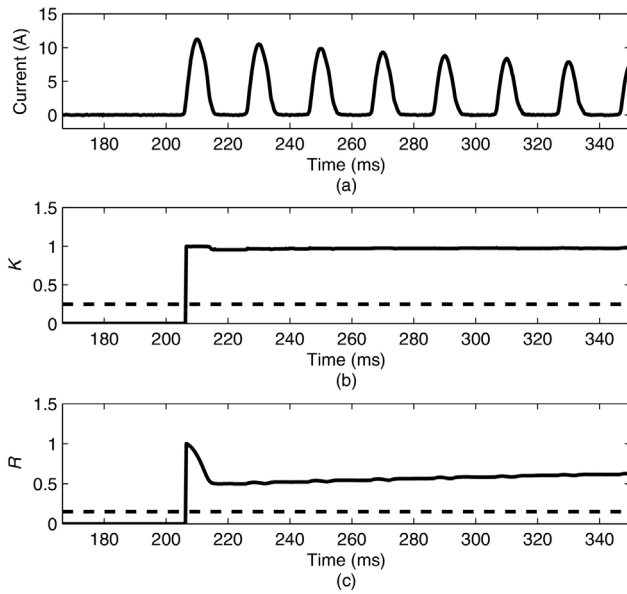


Fig. 3. Transformer energizing at positive zero crossing of phase A voltage: (a) Current waveform, (b) the proposed index, and (c) the ratio of the second harmonic to the fundamental component.

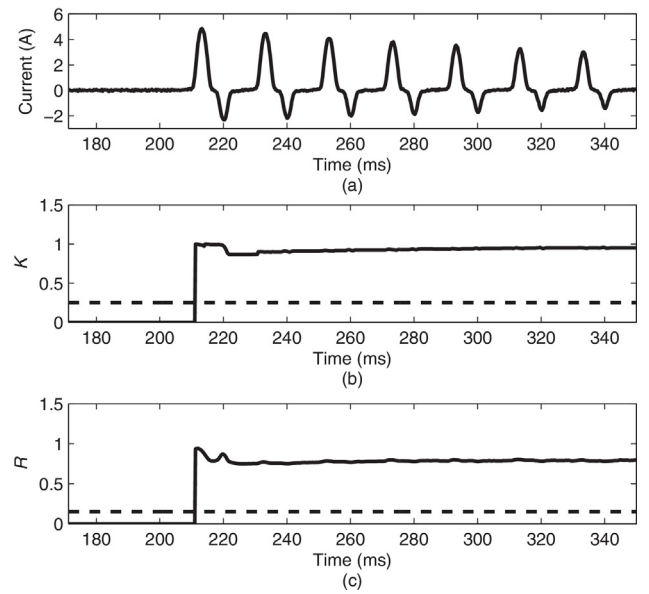


Fig. 4. Transformer energizing at 120° of phase A voltage: (a) Current waveform, (b) the proposed index, and (c) the ratio of the second harmonic to the fundamental component.

3.2. Inrush currents

3.2.1. Effect of switching instants

Fig. 3(a) shows the waveform of the inrush current where the transformer is switched on at the zero crossing of the corresponding voltage. The dynamic changes of the proposed index K and the ratio of the second harmonic to the fundamental component R are depicted in solid lines in Fig. 3(b) and (c), respectively. The dashed lines in the figures are respectively the corresponding thresholds. We can see that both of the two indices are much higher than their corresponding thresholds. The proposed index remains about 1 after the switching whereas the ratio of the second harmonic to the fundamental component has an obvious reduction in the first cycle after switching.

Fig. 4(a) shows the waveform of the inrush current where the transformer is switched on at 120° of the phase A voltage. In this case phase B has the maximum inrush current and the polarities of the currents of phase A and phase C are the same, which leads to a symmetrical differential current. As shown in Fig. 4(b) and (c), both of the proposed index and the second harmonic ratio are larger than the thresholds and thus both methods can identify the inrush correctly.

3.2.2. Effect of CT saturation

CT saturation is another factor may affect the performance of inrush identification method. Fig. 5(a) shows the waveform of the distorted inrush current due to CT saturation. From Fig. 5(b) and (c), we can see that CT saturation results in fluctuation in both of the proposed index and second harmonic ratio, however, both of the two indices are still much larger than their corresponding thresholds in this case.

3.2.3. Effect of remanence

The residual magnetism in the transformer core may exist due to previous switching operation of the transformer, which can increase the peak value of the inrush current significantly. Fig. 6(a) shows the waveform of an inrush current where the transformer is energized with 80% remanence at the zero crossing of voltage. As depicted in Fig. 6(b), the proposed index has a small reduction during the first cycle, however, it stays much larger than the threshold.

In contrast, the ratio of the second harmonic to the fundamental frequency component decreases significantly due to the high remanence. The ratio drops to 14.2% in the first cycle after the switching, which is smaller than the threshold and thus the second harmonic restraint method falsely identify the inrush current as a fault.

In order to comprehensively study the effects of the three factors on the performance on the proposed method, 105 cases of inrush currents generated under different switching conditions were simulated by using the Multiple Run component in PSCAD/EMTDC. The energizing time of transformer varies between 200 ms and 220 ms, corresponding to 0–360° of phase A voltage, as shown in Fig. 7(a). The controlled DC sources are respectively adjusted to generate remanence as 0%, 20%, 40%, 60% and 80% of the peak flux generated at rated voltage, as shown in Fig. 7(b).

For the 105 cases, the minimum values of the proposed index denoted by $K_{\min}$ and the ratio of the second harmonic to the

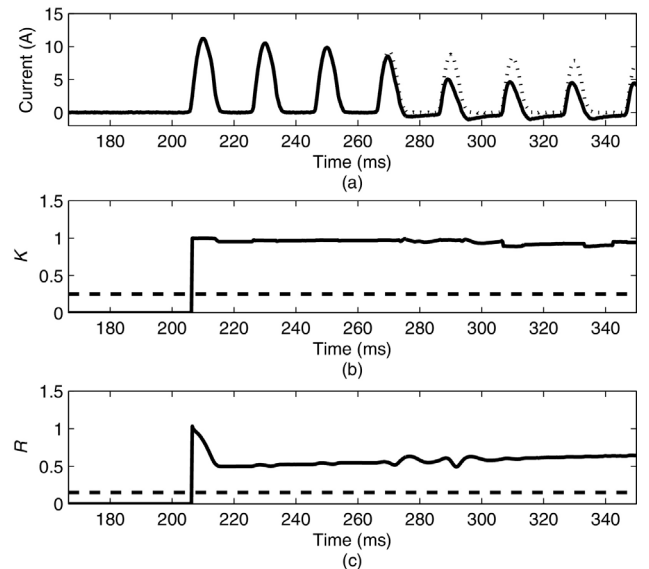


Fig. 5. Inrush current distorted by CT saturation: (a) Current waveform, (b) the proposed index, and (c) the ratio of the second harmonic to the fundamental component.

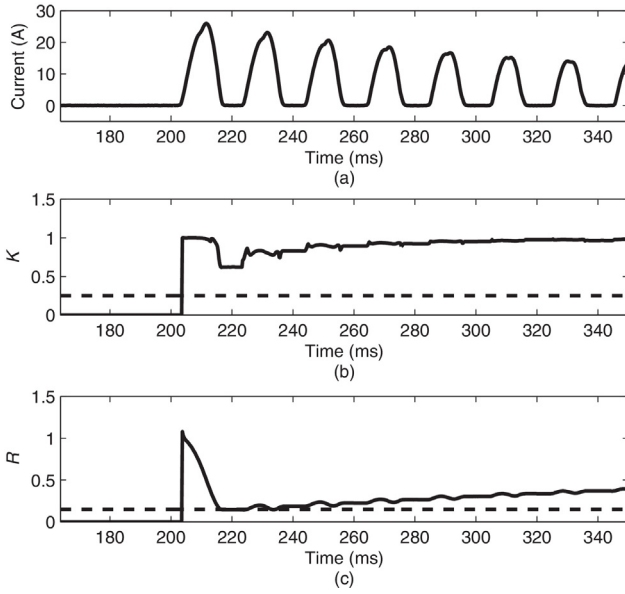


Fig. 6. Transformer energizing with 80% remanence: (a) Current waveform, (b) the proposed index, and (c) the ratio of the second harmonic to the fundamental component.

fundamental component denoted by $R_{\min}$ are depicted in Fig. 8(a) and (b), respectively. It can be seen that the effect of switching instants on the performance of the proposed index is limited when the level of remanence is not high. Although CT saturation and high level of remanence lead to reductions in the values of $K_{\min}$, the proposed method correctly identified the inrush currents and reliably restrained the relay in all cases. In contrast, the performance of the second harmonic restraint method is strongly affected by the switching instants regardless of the level of remanence. In some cases, the values of $R_{\min}$ are smaller than the threshold due to the CT saturation and the high level of remanence.

As summarized in Table 2, without considering CT saturation, $K_{\min}$ varies between 0.3171 and 0.9858 while $R_{\min}$ varies between 0.1385 and 0.7669. There are 3 cases are falsely identified as internal faults by the second harmonic restraint method. Considering CT saturation, $K_{\min}$ varies between 0.3146 and 0.9155 while $R_{\min}$ varies between 0.038 and 0.7686. There are 16 cases are identified by the second harmonic restraint method as internal faults. The

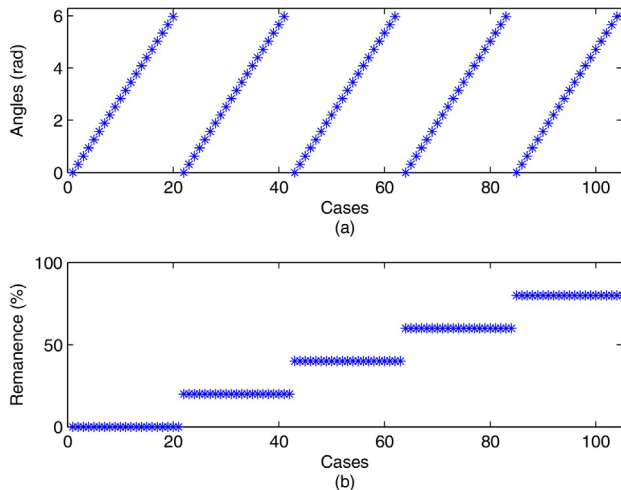


Fig. 7. Conditions for transformer energizing for the 105 cases of inrush currents: (a) Switching angles, and (b) the levels of remanence.

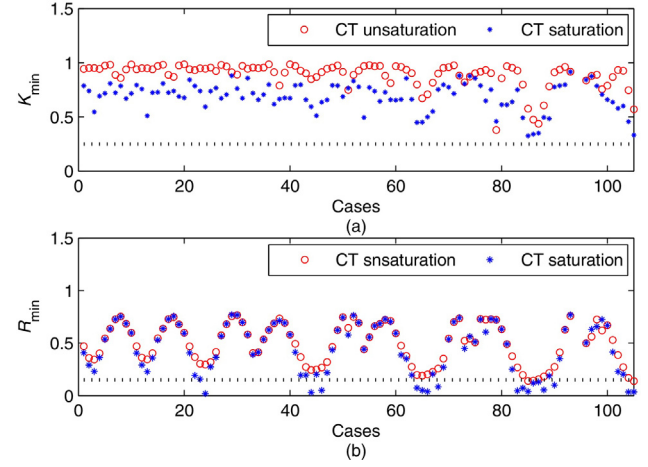


Fig. 8. Minimum values of different indices for the 105 cases of inrush currents: (a) The proposed index, (b) the ratio of the second harmonic to the fundamental component.

Table 2

Simulation results for inrush currents under different conditions.

Method	CT unsaturation		CT saturation	
	Range of $K_{\min}$	Errors	Range of $R_{\min}$	Errors
Proposed method	[0.3171, 0.9858]	0/105	[0.3146, 0.9155]	0/105
Second harmonic	[0.1385, 0.7669]	3/105	[0.0380, 0.7686]	16/105

results verify that the proposed method is more reliable than the conventional second harmonic restraint method, even though the threshold K_{set} is higher than R_{set} .

3.3. Internal fault currents

Fig. 9(a) shows the differential current of the case of a signal phase-to-ground fault in the power transformer. It can be seen in Fig. 9(b) and (c) that, without considering CT saturation, the proposed index and the second harmonic ratio fall below corresponding thresholds at almost the same time in less than a cycle

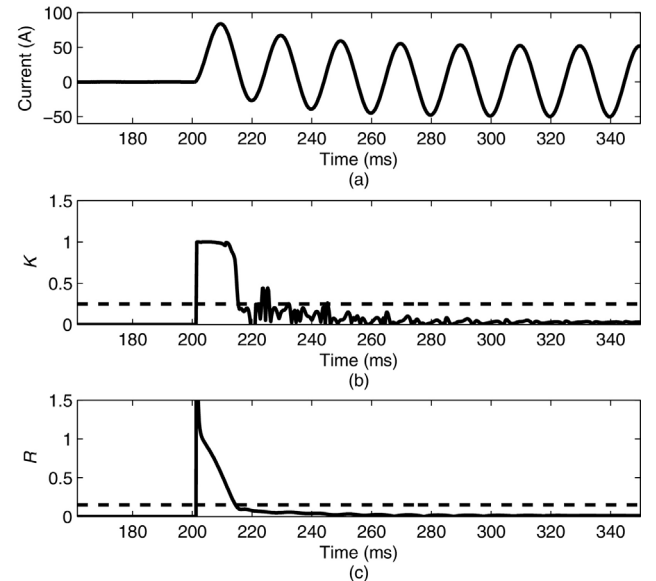


Fig. 9. Internal fault current without CT saturation: (a) Current waveform, (b) the proposed index, and (c) the ratio of the second harmonic to the fundamental component.

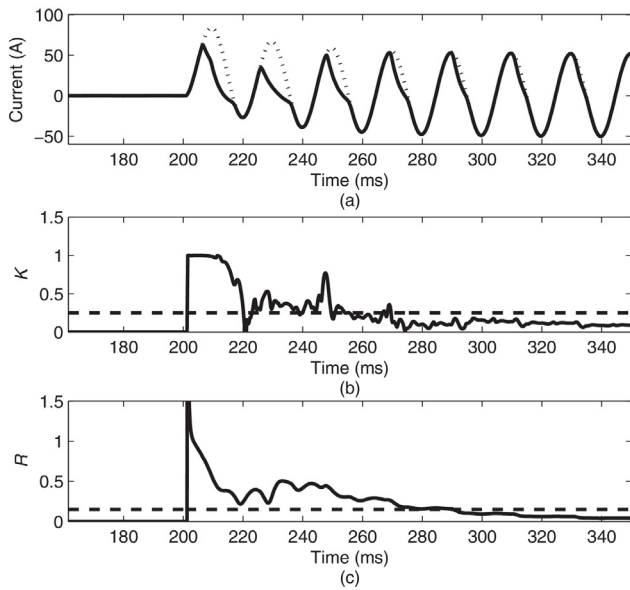


Fig. 10. Internal fault current with CT saturation: (a) Current waveform, (b) the proposed index, and (c) the ratio of the second harmonic to the fundamental component.

after fault inception. However, both of the values of the indices may increase due to CT saturation, which leads to a longer delay for the operation of differential relays. Fig. 10 show the result of the same case when considering CT saturation. It can be seen that the proposed index falls below 0.25 at about a cycle after fault inception. On the other hand, the ratio of second harmonic to fundamental frequency component stays above 0.15 for about 4 cycles.

420 cases in total were conducted using the Multiple Run component in PSCAD/EMTDC to simulate internal faults under different conditions with 4 different fault types, 5 different fault locations and 21 different fault inception angles. The fault types include single-phase-to-ground fault (AG), two-phase fault (AB), two-phase-to-ground fault (ABG) and three-phase-to-ground fault (ABCG). The fault inception angles vary between 0 and 360°, at a step of 18°. The fault locations in the winding partition are set to 0%, 20%, 40%, 60%, 80%, respectively.

Table 3 summarizes the time delays for the two methods to detect all the internal faults with respect to different fault types. As mentioned above, each type of fault was applied at 5 different fault locations and 21 fault inception angles. In Table 3, t_{mean} , t_{max} and t_{min} are the average, maximum and minimum values of time delays for 105 (5 × 21) cases, respectively. It can be seen that without considering CT saturation the time delays for the two methods to identify the internal faults are very close.

Table 3
Time delays for internal faults with respect to different fault types (ms).

Fault type	Method	CT unsaturation			CT saturation		
		t_{mean}	t_{max}	t_{min}	t_{mean}	t_{max}	t_{min}
AG	Proposed method	15.24	21.25	6.00	15.32	21.25	6.25
	Second harmonic	16.45	20.50	11.25	16.43	20.50	11.00
ABG	Proposed method	16.65	21.00	13.50	27.80	91.00	13.75
	Second harmonic	15.54	18.50	12.25	37.33	113.25	15.25
ABG	Proposed method	16.56	21.00	13.75	27.18	92.75	14.50
	Second harmonic	15.48	18.50	12.50	36.81	96.00	15.25
ABCG	Proposed method	17.00	21.00	14.00	20.53	75.25	11.25
	Second harmonic	15.58	18.50	12.50	23.76	75.25	13.75

$K_{\text{set}} = 0.25$, $R_{\text{set}} = 0.15$.

However, if the differential currents are distorted by CT saturation, the time delays of the proposed method are much smaller than those of the second harmonic restraint method, especially for the two-phase-to-ground faults and two-phase faults.

Table 4 summarizes the time delays for the two methods to detect all the internal faults with respect to different fault locations in winding partition. At each fault location, 4 different fault types and 21 different fault inception angles were applied respectively. Therefore, at each fault location, 84 (4 × 21) cases were simulated. It can be seen that when considering CT saturation, the time delays of the proposed method are much smaller than that of the second harmonic restraint method, especially for the cases where the fault locations are close to the two ends of the winding.

It should be noted that the results in these two tables are obtained with $K_{\text{set}} = 0.25$, $R_{\text{set}} = 0.15$. With this setting, the proposed method can correctly identify all the cases of inrush conditions while the second harmonic restraint method falsely identify some of the cases, as indicated in Table 2. This is to say, the reliability of the proposed method is much higher than that of the second harmonic restraint method. In fact, the selection of threshold is a trade-off between the response speed and reliability. The higher the threshold, the faster the method can respond to internal faults but the lower the reliability, and vice versa. As indicated in Table 2, the threshold of the second harmonic restraint method R_{set} should be set to a much lower threshold, like 0.03, in order to avoid the malfunction of differential protection under some of the inrush conditions, which however will significantly increase the time delays. Table 5 shows the time delays, obtained with $K_{\text{set}} = 0.25$ and $R_{\text{set}} = 0.03$, for the 420 internal faults with respect to different fault locations. It is shown that the proposed method is faster than the second harmonic restraint method no matter whether the CTs are saturated or not. In this respect, the proposed method is better than the second harmonic restraint method in terms of both reliability and response speed.

3.4. Internal fault current and inrush current occur simultaneously

When the transformer is switched on with an internal fault, inrush current and fault current occur simultaneously. The conventional second harmonic restraint method may restrain the relay for a long time in this situation.

In order to evaluate the performance of the proposed method under this situation, 420 cases were simulated. The power transformer T2 were energized with the above mentioned 4 different internal fault types at 5 different locations in the winding partitions and 21 different angles of the phase A voltage, respectively. Table 6 summarizes the time delays for the two methods to correctly identify the internal faults. By comparing the data in Tables 4 and 6,

Table 4

Time delays for internal faults detection with respect to different fault locations (ms).

Fault location	Method	CT unsaturation			CT saturation		
		t_{mean}	t_{max}	t_{min}	t_{mean}	t_{max}	t_{min}
0%	Proposed method	15.91	18.50	11.25	17.99	73.25	11.25
	Second harmonic	15.68	18.75	12.75	36.48	91.25	13.75
20%	Proposed method	15.99	18.75	11.00	17.16	22.75	7.75
	Second harmonic	15.60	18.50	12.75	20.28	30.25	14.25
40%	Proposed method	15.80	18.50	11.75	16.81	22.00	11.75
	Second harmonic	15.55	18.50	12.50	19.32	29.50	14.00
60%	Proposed method	15.51	18.50	11.50	16.49	22.00	11.00
	Second harmonic	15.23	18.50	12.50	18.93	29.25	13.25
80%	Proposed method	14.97	18.25	6.25	20.81	75.25	6.50
	Second harmonic	14.89	18.25	11.50	24.11	92.25	11.50

 $K_{\text{set}} = 0.25$, $R_{\text{set}} = 0.15$.**Table 5**

Time delays for internal faults detection with respect to different fault locations (ms).

Fault location	Method	CT unsaturation			CT saturation		
		t_{mean}	t_{max}	t_{min}	t_{mean}	t_{max}	t_{min}
0%	Proposed method	15.91	18.50	11.25	17.99	73.25	11.25
	Second harmonic	30.66	45.75	17.50	86.96	171.25	16.00
20%	Proposed method	15.99	18.75	11.00	17.16	22.75	7.75
	Second harmonic	29.99	46.75	17.25	89.78	174.50	17.25
40%	Proposed method	15.80	18.50	11.75	16.81	22.00	11.75
	Second harmonic	29.37	45.75	16.75	82.02	179.50	16.75
60%	Proposed method	15.51	18.50	11.50	16.49	22.00	11.00
	Second harmonic	28.51	45.25	15.00	55.37	168.75	15.00
80%	Proposed method	14.97	18.25	6.25	20.81	75.25	6.50
	Second harmonic	26.25	45.00	12.50	65.55	173.25	12.50

 $K_{\text{set}} = 0.25$, $R_{\text{set}} = 0.03$.

we can see that when the fault current is contaminated by inrush, both the time delays of the two methods increase. However, the response speed of the proposed method is much faster than that of the second harmonic restraint method, especially when the fault locations are close to the two ends of windings.

Fig. 11(a) shows an example where the transformer is energized with a three phase fault located at 80% partition of winding. As shown in Fig. 11(c), the ratio of the second harmonic to fundamental frequency component in phase A decreases gradually and is still larger than 0.15 at 150 ms after the energization of transformer. In contrast, the differential current was correctly identified as internal fault current by the proposed method in less than 20 ms.

4. Experimental studies

The effectiveness of the proposed index is further examined by using recorded data generated by experiments on a three-phase 800/220 V, 15 kVA, Yd11 connected transformer in laboratory. A total of 28 cases with different inrush currents and internal faults were recorded by a relay fault recorder at a sampling rate of 10 kHz. After the data was imported into MATLAB, differential currents were calculated considering the mismatch of the ratios of the CTs on the primary side and secondary side of the transformer.

Fig. 12(a) shows a case of inrush current where the transformer was energized at 166° of voltage. As shown in Fig. 12(b), the

Table 6

Time delays for the Conditions of inrush and internal fault current occur simultaneously with respect to different fault locations (ms).

Fault location	Method	CT unsaturation			CT saturation		
		t_{mean}	t_{max}	t_{min}	t_{mean}	t_{max}	t_{min}
0%	Proposed method	22.88	131.75	13.50	20.60	109.75	11.25
	Second harmonic	44.63	270.75	12.50	55.15	250.50	14.00
20%	Proposed method	16.40	19.00	14.00	19.58	72.50	14.00
	Second harmonic	27.82	272.75	12.50	34.09	187.25	14.00
40%	Proposed method	16.15	18.75	14.00	18.90	79.00	14.00
	Second harmonic	15.38	17.00	12.75	22.18	70.25	14.50
60%	Proposed method	16.11	18.75	14.00	17.52	22.75	14.00
	Second harmonic	15.39	17.25	12.75	20.96	30.00	14.50
80%	Proposed method	15.94	18.50	13.50	24.23	112.00	13.50
	Second harmonic	59.56	272.25	12.25	43.90	195.25	14.00

 $K_{\text{set}} = 0.25$, $R_{\text{set}} = 0.15$.

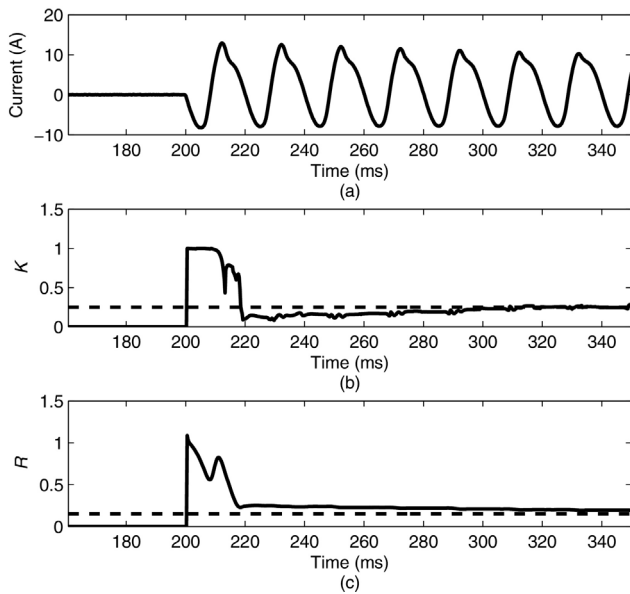


Fig. 11. Internal fault and inrush current occurring simultaneously: (a) Current waveform, (b) the proposed index, and (c) the ratio of the second harmonic to the fundamental component.

proposed index is always larger than 0.5. Therefore, the proposed index correctly identified the differential current as an inrush.

Fig. 13 shows the case where the experimental transformer was switched on with a turn-to-turn fault. Although the fault current is contaminated by inrush current, it can still be recognized as internal fault by the proposed method and the time delay is only 13.25 ms. On the other hand, the time delay of the second harmonic restraint method is 27.25 ms.

Both the proposed method and the second harmonic restraint method acted correctly in all the 28 cases, which demonstrated that both methods can be applied to the protection of the low voltage power transformer where the sampling rate is different from that used in the simulation studies.

However, if the experimental studies were performed on the high voltage power transformer, the ratios of second-harmonic

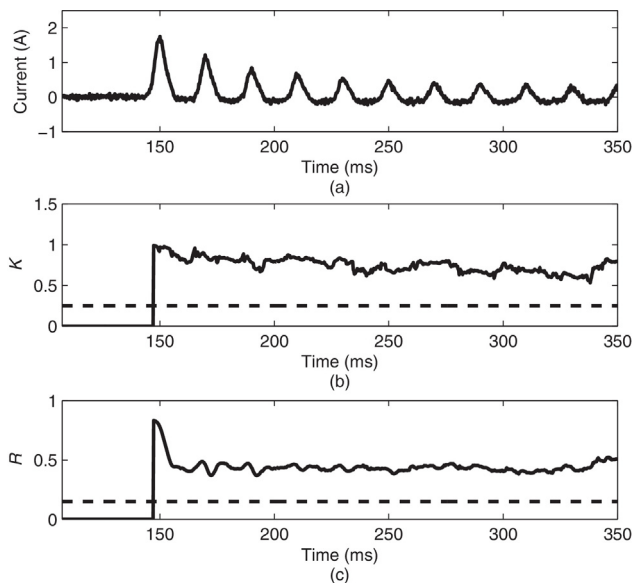


Fig. 12. The identification results of an inrush current in laboratory experiment: (a) Current waveform, (b) the proposed index, and (c) the ratio of the second harmonic to the fundamental component.

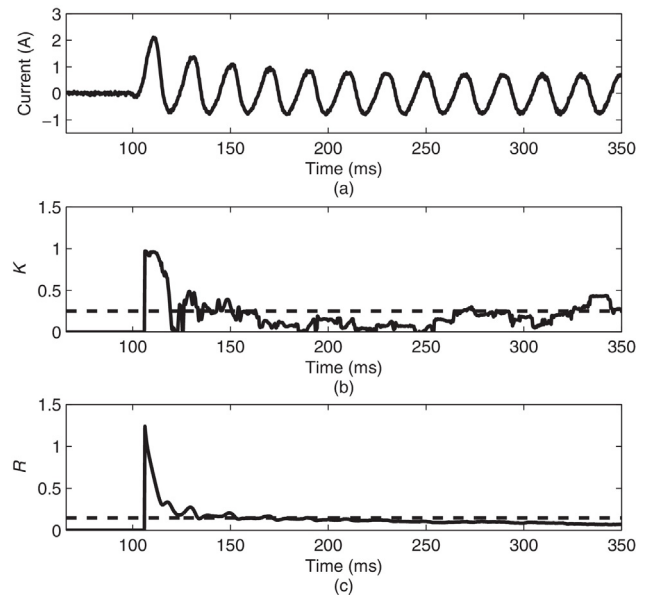


Fig. 13. The identification results of energizing transformer on an internal fault in laboratory experiment: (a) Current waveform, (b) the proposed index, (c) the ratio of the second harmonic to the fundamental component.

component in the inrush current can be below 10% under some conditions [31]. This indicates that the threshold of the second harmonic restraint method, R_{set} , need to be set to a lower value to avoid mal-operation of the differential relay. Nevertheless, the ratio of the second-harmonic in fault current is relatively high in the beginning of fault inception for some slight faults such that the time delay for the second harmonic restraint method to identify the faults will be longer.

On the other hand, the proposed method combines three higher-order-statistics-based indices which measure the characteristics of the distributions of the sample points of the differential current from different perspectives. As shown in the simulation results, it can correctly identify the inrush current where the ratio of second harmonic component is quite low. Therefore, it should be able to work well if the experimental studies were performed on a high voltage power transformer.

5. Conclusion

This paper has proposed a higher-order statistics based method for the discrimination of magnetizing inrush and internal fault. Extensive simulation studies and experimental tests have demonstrated that the proposed method:

- (1) can reliably restrain differential relays for inrush currents occurring under various conditions, including CT saturation and energizing transformer with high level of remanence where the conventional methods may malfunction.
- (2) can release differential relays much faster than the second harmonic restraint method for the cases where the internal fault current is distorted by CT saturation or contaminated with an inrush current.

Acknowledgement

This work was supported by the Guangdong Innovative Research Team Program (No. 201001N0104744201) and by China Postdoctoral Science Foundation (No. 2015M582383).

References

- [1] T.S. Sidhu, M.S. Sachdev, Online identification of magnetizing inrush and internal faults in three-phase transformers, *IEEE Trans. Power Deliv.* 7 (4) (1992) 1885–1891.
- [2] H. Dashti, M. Davarpanah, M. Sanaye-Pasand, H. Lesani, Discriminating transformer large inrush currents from fault currents, *Int. J. Electr. Power Energy Syst.* 75 (2016) 74–82.
- [3] H. Zhang, P. Liu, O.P. Malik, A new scheme for inrush identification in transformer protection, *Electr. Power Syst. Res.* 63 (2) (2002) 81–86.
- [4] M.E.H. Golshan, M. Saghaian-Nejad, A. Saha, H. Samet, A new method for recognizing internal faults from inrush current conditions in digital differential protection of power transformers, *Electr. Power Syst. Res.* 71 (1) (2004) 61–71.
- [5] A. Hosny, V.K. Sood, Transformer differential protection with phase angle difference based inrush restraint, *Electr. Power Syst. Res.* 115 (2014) 57–64.
- [6] K. Yabe, Power differential method for discrimination between fault and magnetizing inrush current in transformers, *IEEE Trans. Power Deliv.* 12 (3) (1997) 1109–1118.
- [7] A. Hooshyar, M. Sanaye-Pasand, S. Afsharnia, M. Davarpanah, B.M. Ebrahimi, Time-domain analysis of differential power signal to detect magnetizing inrush in power transformers, *IEEE Trans. Power Deliv.* 27 (3) (2012) 1394–1404.
- [8] S.R. Huang, H.T. Chen, C.C. Wu, C.Y. Guan, C. Cheng, Distinguishing internal winding faults from inrush currents in power transformers using Jiles–Atherton model parameters based on correlation coefficient, *IEEE Trans. Power Deliv.* 27 (2) (2012) 548–553.
- [9] J. Ma, Z. Wang, Q. Yang, Y. Liu, Identifying transformer inrush current based on normalized grille curve, *IEEE Trans. Power Deliv.* 26 (2) (2011) 588–595.
- [10] R.J.N. Alencar, U.H. Bezerra, A method to identify inrush currents in power transformers protection based on the differential current gradient, *Electr. Power Syst. Res.* 111 (6) (2014) 78–84.
- [11] X.N. Lin, P. Liu, O.P. Malik, Studies for identification of the inrush based on improved correlation algorithm, *IEEE Trans. Power Deliv.* 17 (4) (2002) 901–907.
- [12] H. Zhang, J.F. Wen, P. Liu, O.P. Malik, Discrimination between fault and magnetizing inrush current in transformers using short-time correlation transform, *Int. J. Electric. Power Energy Syst.* 24 (7) (2002) 557–562.
- [13] O.A.S. Youssef, A wavelet-based technique for discrimination between faults and magnetizing inrush currents in transformers, *IEEE Trans. Power Deliv.* 18 (1) (2003) 170–176.
- [14] A.R. Sedighi, M.R. Haghifam, Detection of inrush current in distribution transformer using wavelet transform, *Int. J. Electric. Power Energy Syst.* 27 (5–6) (2005) 361–370.
- [15] J. Faiz, S. Lotfi-Fard, A novel wavelet-based algorithm for discrimination of internal faults from magnetizing inrush currents in power transformers, *IEEE Trans. Power Deliv.* 21 (4) (2006) 1989–1996.
- [16] A.M. Gaouda, M.M.A. Salama, Dsp wavelet-based tool for monitoring transformer inrush currents and internal faults, *IEEE Trans. Power Deliv.* 25 (3) (2010) 1258–1267.
- [17] A.A.H. Eldin, M.A. Refaey, A novel algorithm for discrimination between inrush current and internal faults in power transformer differential protection based on discrete wavelet transform, *Electr. Power Syst. Res.* 81 (1) (2011) 19–24.
- [18] B. Vahidi, N. Ghaffarzadeh, S.H. Hosseini, A wavelet-based method to discriminate internal faults from inrush currents using correlation coefficient, *Int. J. Electric. Power Energy Syst.* 32 (7) (2010) 788–793.
- [19] M. Rasoulpoor, M. Banejad, A correlation based method for discrimination between inrush and short circuit currents in differential protection of power transformer using discrete wavelet transform: theory, simulation and experimental validation, *Int. J. Electric. Power Energy Syst.* 51 (10) (2013) 168–177.
- [20] P. Sun, J.F. Zhang, D.J. Zhang, Q.H. Wu, S. Potts, Morphological identification of transformer magnetising inrushcurrent, *Electron. Lett.* 38 (9) (2002) 437–438.
- [21] Z. Lu, W.H. Tang, T.Y. Ji, Q.H. Wu, A morphological scheme for inrush identification in transformer protection, *IEEE Trans. Power Deliv.* 24 (2) (2009) 560–568.
- [22] W. Wu, T. Ji, M. Li, Q. Wu, Using mathematical morphology to discriminate between internal fault and inrush current of transformers, *IET Gener. Transm. Distrib.* 10 (1) (2015) 73–80.
- [23] L.G. Perez, A.J. Flechsig, J.L. Meador, Z. Obradovic, Training an artificial neural network to discriminate between magnetizing inrush and internal faults, *IEEE Trans. Power Deliv.* 9 (1) (1994) 434–441.
- [24] M. Yazdani-Asrami, M. Taghipour-Gorjilaie, S.M. Razavi, S.A. Gholamian, A novel intelligent protection system for power transformers considering possible electrical faults, inrush current, ct saturation and over-excitation, *Int. J. Electric. Power Energy Syst.* 64 (2015) 1129–1140.
- [25] P.L. Mao, R.K. Aggarwal, A wavelet transform based decision making logic method for discrimination between internal faults and inrush currents in power transformers, *Int. J. Electric. Power Energy Syst.* 22 (6) (2000) 389–395.
- [26] S.R. Samantaray, P.K. Dash, Decision tree based discrimination between inrush currents and internal faults in power transformer, *Int. J. Electric. Power Energy Syst.* 33 (33) (2011) 1043–1048.
- [27] X. Ma, J. Shi, A new method for discrimination between fault and magnetizing inrush current using HMM, *Electr. Power Syst. Res.* 56 (1) (2000) 43–49.
- [28] L.L. Zhang, Q.H. Wu, T.Y. Ji, L. Jiang, Skewness-based differential protection scheme for EHV/UHV transmission lines, *IEEE Trans. Power Deliv.* 29 (3) (2014) 1518–1520.
- [29] J.M. Mendel, Tutorial on higher-order statistics (spectra) in signal processing and system theory: theoretical results and some applications, *Proc. IEEE* 79 (3) (1991) 278–305.
- [30] Manitoba HVDC Research Centre Inc., Applications of PSCAD/EMTDC, Manitoba HVDC Research Centre Inc., Winnipeg, MB, 2007.
- [31] X. Lin, J. Huang, L. Zeng, Z.Q. Bo, Analysis of electromagnetic transient and adaptability of second-harmonic restraint based differential protection of UHV power transformer, *IEEE Trans. Power Deliv.* 25 (4) (2010) 2299–2307.